ANALYZING MOVIE REVENUE

MOTIVATION AND RESEARCH QUESTION

- Research question: How do movie characteristics influence the financial success of films?
- Why it matters: Knowing drivers of revenue supports budgeting, marketing strategy and risk reduction.
- Goal: Quantify which factors best predict revenue (e.g., budget/genre/popularity/ratings).
- Prior evidence: Past studies link revenue to budget, genre, and audience response (word-of-mouth/ratings).

DATA COLLECTION

- Source: TMDb 5000 Movies (Kaggle), from The Movie Database (TMDb). (Kaggle, n.d.)
- Collection: Aggregated movie metadata via TMDb platform/API (user + partner contributions).
- Sample: Revenue > 0 → n = 3,228; revenue has no missing values.
- Why appropriate: Contains revenue + key predictors (budget, runtime, genre, popularity, vote metrics).
- Trust & caveats: Large, structured dataset, but may be biased toward mainstream/Hollywood/English and incomplete for niche/older films.

METHODS OF ANALYSIS

Data Cleaning and Variable Construction

- Load dataset
- **data cleaning and variable creation (tidyverse):** Remove movies with non-positive budget
- **Create new variables for model:**

Initial Model Fit and Diagnostic Assessment

Fit Preliminary Model (10 .predictors)

- $R^2 \approx 0.72$, Adj $R^2 \approx 0.71$,
- F-test $p < 0.001$

Findings:

- heteroskedastic Residuals
- Q-Q Plot Heavy-tailed
- Several **high-leverage** blockbuster **outliers**

Transformation Testing and Model Refinement

Square-root model → rejected

- Create sqrt_revenue, sqrt_budget, sqrt_vote_count

Result: Variance unstable

Box–Cox model → rejected

- $\lambda = 0.3454$
- MASS::boxcox

Result: extreme standardized residuals

Final Model Selection and Justification

log10 transformed model was chosen:

- log_revenue, log_budget, log_vote_count
- **Adjusted R^2** : 0.6505
- **AIC:** 5123.1 **BIC**: 5402.7
- Residual Std. Error: 0.531

Log-linear model has best AIC/BIC and results

Final Diagnostic Verification

Check all assumptions on transformed model

- Residuals vs Fitted
- Q–Q
- Scale–Location: √|residuals|
- Leverage: Cook's distance
- R diagnostic plots (plot.lm)

CONCLUSION

- The log-linear model best explained revenue variation, with the lowest **AIC (5123.1)** and **BIC (5402.7)**, making it the strongest model for identifying revenue drivers.
- **Budget** strongly increased revenue → production scale is a major factor for revenue
- **Audience reach (vote_count)**, was influential → films with wider engagement earns more.
- **Release year** had a small negative effect → newer films have had less time to accumulate revenue.
- **Vote_average** showed a negative association → ratings alone do not drive revenue once visibility and budget are controlled .
- **Genre** has effects, with **Documentary** showing a strong positive association → genre differences affect financial performance.

RESULTS

Residuals vs Fitted

- Residuals mostly −2 to +2; variance more stable.
- Minimal curvature; heteroskedasticity reduced.

Q–Q Plot

- Central points follow normal line.
- Only mild upper-tail deviations (~+5 to +6).

Scale–Location

- √|residuals| mainly 0–1.2; trend nearly flat.
- Variance clearly stabilized vs raw model.

Residuals vs Leverage

- Most leverage < 0.05; max ≈ 0.25.
- No points exceed Cook's distance.

Table 1. Top significant predictors from the final log-linear model.

Predictor	Estimate	Std Error	t-value	p-value	CI-Lower	CI-Upper	Sig.
Intercept	27.68	1.784	15.518	< 2e-16	24.180	31.180	***
log_budget	0.66	0.02029	32.37	< 2e-16	0.617	0.697	***
log_vote_count	0.77	0.02421	31.611	< 2e-16	0.718	0.813	***
release_year	−0.01326	0.0008915	−14.872	< 2e-16	−0.015	−0.012	***
vote_average	−0.05545	0.0155	−3.577	0.000352	−0.086	−0.025	***
genreDocumentary	0.76	0.1286	5.944	3.08e-09	0.511	1.017	***

FIGURE 1

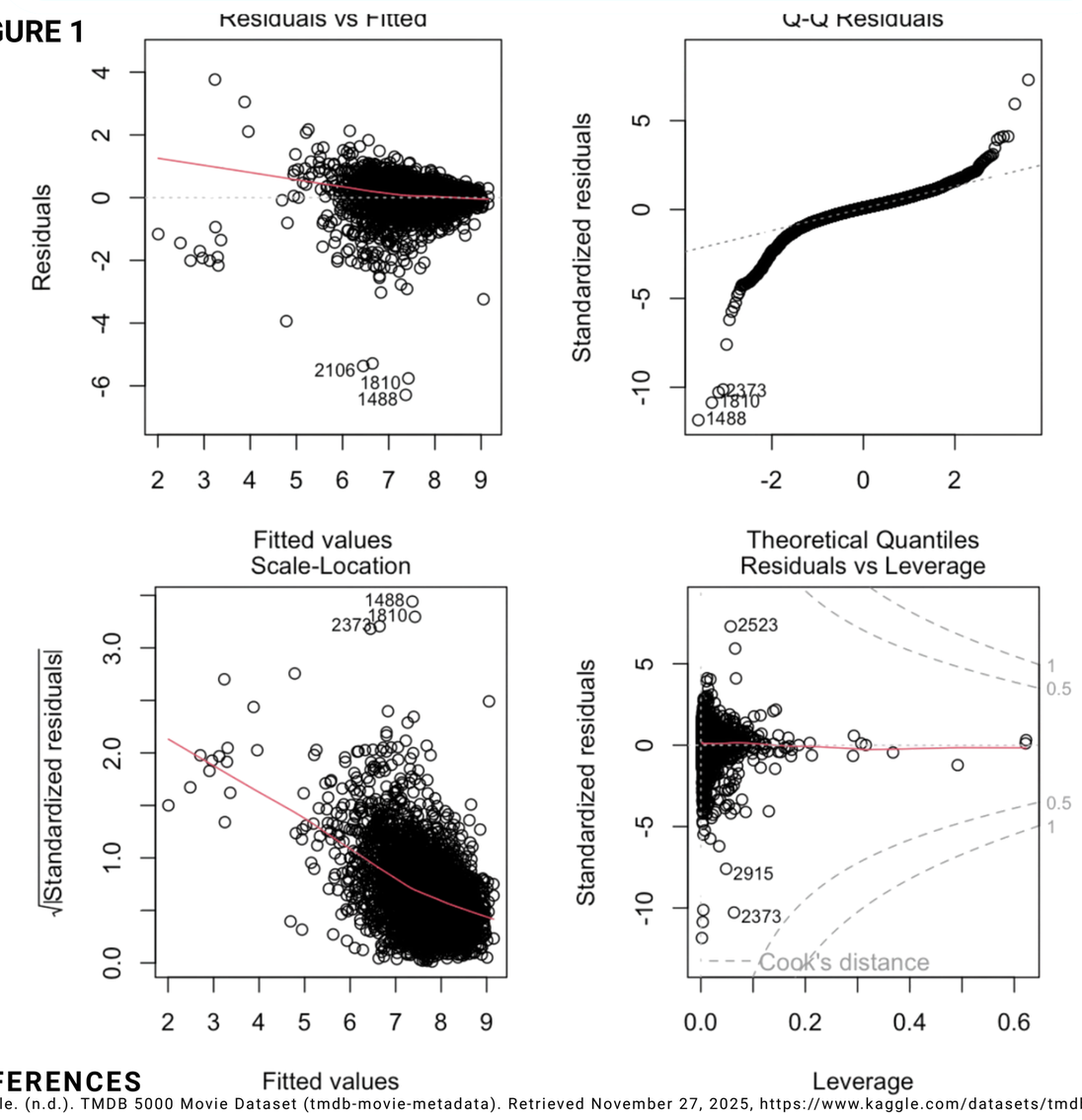

Figure 1. Diagnostics for the log-linear model showing improved variance, normality, and low leverage.

LIMITATIONS

- TMDb is user/partner contributed, so records can be incomplete/inconsistent for older/smaller/niche films and may reflect mainstream/Hollywood/English audience patterns. "popularity" may reflect TMDb users rather than the general public.
- Future work could incorporate more complete revenue data sources and more variables relating to revenue to increase predictive accuracy.

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